Hasini Lavudiya

ch23btech11027@iith.ac.in | lavudiyahasini@gmail.com | +91 7093992113 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.32
Aaditya Junior College	Class XII (Telangana State borad)	2023	67.00%
Paramita IIT School	Class X (SSC)	2020	10.00%

SCHOLASTIC ACHIEVEMENTS

- Secured admission to **IIT Hyderabad** in Chemical Engineering through JEE Advanced.
- Winner of multiple dance competitions at the school level.
- Awarded best sportsperson for outstanding performance across multiple sports disciplines at the school level.

RELEVANT COURSES

- Artificial Intelligence
- Computer Vision in Chemical and Life Science
- Optimization Techniques-I
- Principles of Game Theory
- Molecular Modelling of Catalytic Reactions

SKILLS

Programming:

- Python, C/C++

Libraries/Tools:

- NumPy, Pandas, Matplotlib, OpenCV, Jupyter Notebook, Git/GitHub

AI/ML:

- Machine Learning, Neural Networks, Restricted Boltzmann Machines, Energy-Based Models

Computational Tools:

- MATLAB, ASE, VASP, Material Studio

PROJECTS

Multi-Objective Optimization of Airfoil Performance | *Python, TensorFlow, Scikit-learn, Pymoo*

- Developed an **ANN surrogate model** to predict airfoil lift and drag coefficients from aerodynamic parameters.
- Applied **NSGA-II multi-objective optimization** to maximize lift while minimizing drag.
- Generated and analyzed the **Pareto front** to identify optimal lift-drag trade-offs.
- https://github.com/H49686/airfoil-optimization

Linear and Quadratic Discriminant Analysis for Classification | *Python, Scikit-learn, ML*

- Implemented **LDA and QDA** on the UCI Raisin dataset using preprocessing, standardization and 10-fold cross-validation.
- Evaluated classification performance using **Accuracy, Precision, Recall, F1-score and ROC-AUC**, achieving 88.3% test accuracy with tuned LDA.
- https://github.com/H49686/Raisin-LDA-QDA

PID and Model Predictive Control for Process Optimization | *Julia, JuMP, Ipopt*

- Developed a multivariable process-control simulation for moisture, temperature and light-intensity regulation.
- Implemented **PID and MPC** using feedback control, system dynamics and constrained optimization.
- Evaluated controller performance using **PCI tracking and time-series analysis**.
- https://github.com/H49686/Basil-PID-MPC-Control

EXTRACURRICULAR

- Dance
- Performing Arts
- Drawing
- Reading
- Logical Games

KEY STRENGTHS

- Communication
- Initiative
- Adaptability
- Time Management
- Attention to Detail
- Growth Mindset